

Specification: Statistical Analysis

1 Objective

Run statistical analysis on the unified evaluation dataset to reproduce the paper's findings.

2 Input

File	Format	Description
data/unified_evaluations.csv	CSV	Combined evaluations from all judges and models

3 Required Columns

The input CSV must contain these columns:

Column	Type	Description
provider	string	Model name (e.g., openai, anthropic, deepseek)
condition	string	Culture_5, Culture_Expert, Culture_11, Large_Baseline
problem_id	integer	0-99
Novelty	float	1-5
Usefulness	float	1-5
Flexibility	float	1-5
Elaboration	float	1-5
Cultural Sensitivity	float	1-5

*Note: Based on our inter-rater reliability analysis, **Novelty** was the only dimension with good agreement among judges ($\kappa = 0.68$). We recommend focusing on Novelty for primary analysis.*

4 Statistical Outputs

Running `python analysis/stats.py` generates:

Output File	Contents
results/descriptive_stats.csv	Mean, std, count by provider and condition
results/effect_sizes.csv	Cohen's d with 95% bootstrap CIs (5,000 resamples)
results/anova_results.csv	Two-way ANOVA (Persona $\times$ Model Family)
results/domain_effects.csv	Effect sizes by domain
results/power_analysis.csv	Post-hoc statistical power
results/leave_one_out_sensitivity.csv	Stability analysis
results/economic_analysis_egr.csv	EGR, cost savings, economic value

5 Interpreting Results

Metric	What It Means
Cohen's $d > 0$	Persona improves Novelty
Cohen's $d < 0$	Persona harms Novelty
$ d < 0.2$	Negligible effect
$0.2 \leq d < 0.5$	Small effect
$0.5 \leq d < 0.8$	Medium effect
$ d \geq 0.8$	Large effect
$EGR > 1$	Economically viable
$EGR > 5$	Exceptional economic value
$EGR > 10$	Transformative economic value
$p < 0.05$	Statistically significant (after FDR correction)

6 Domain Mapping

Problem ID ranges map to domains as follows:

Problem IDs	Domain
0-24	Reasoning
25-49	Instruction
50-74	Creative
75-99	General Knowledge

7 Model Classification

Region	Models
Eastern	deepseek, Alibaba (Qwen)
Western	All others (Anthropic, openai, llama, mistral, Google)

8 Decision Rules for Practitioners

Based on pilot results, apply these rules:

9 Notes for Users

- The script assumes your data follows the unified format exactly
- Cost data in the economic analysis is **hardcoded** with 2025 prices; update the `COST_DATA` dictionary if prices change
- Bootstrap iterations are set to 5,000 for robust confidence intervals
- FDR correction (Benjamini-Hochberg) is applied to all multiple comparisons

Condition	Decision
$EGR > 1.5$ AND performance gain $> 5\%$	Deploy immediately
$EGR > 1.5$ BUT performance gain $\leq 5\%$	Deploy for cost-sensitive applications
$EGR \leq 1.5$ BUT performance gain $> 10\%$	Deploy for quality-critical applications
Performance gain $< 0\%$	Reject or conduct deeper analysis